Linear Algebra and Its Applications

Section 6.S: Supplementary Exercises

Detailed worked solutions

Mathjock

Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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6.S Chapter 6 supplementary exercises

For real vectors, $u \cdot v = u^T v$ and $\|u\| = \sqrt{u^T u}$. Orthogonal means perpendicular; orthonormal additionally requires unit length. The orthogonal decomposition $x = p + z$, with $p \in W$ and $z \in W^\perp$, has $p = \text{proj}_W x$. It yields the Pythagorean identity $\|x\|^2 = \|p\|^2 + \|z\|^2$. A least-squares solution is an *input* $\widehat{x}$ minimizing $\|b - Ax\|$; the corresponding closest *output* is $A\widehat{x}$. These two vectors can even have different numbers of coordinates.

Exercise 6.S.1: Nineteen true–false statements about orthogonality

(a) **False.** The zero vector has length zero. All lengths are nonnegative; only nonzero vectors have strictly positive length.

(b) **True.** $\| - v\| = | - 1|\|v\| = \|v\|$.

(c) **True.** This is the Euclidean distance definition.

(d) **False.** The rule is $\|rv\| = |r|\|v\|$. A negative r and nonzero v disprove the stated formula.

(e) **False.** The zero vector is orthogonal to every vector, but a set containing zero is dependent. Two *nonzero* orthogonal vectors are independent.

(f) **True.** $x \cdot (u - v) = x \cdot u - x \cdot v = 0 - 0 = 0$.

(g) **True.** Expanding gives $\|u + v\|^2 = \|u\|^2 + 2u \cdot v + \|v\|^2$; comparison forces $u \cdot v = 0$.

(h) **True.** The analogous expansion with a minus sign gives $\|u - v\|^2 = \|u\|^2 - 2u \cdot v + \|v\|^2$, again forcing zero dot product.

(i) **False.** The projection onto nonzero u is $((y \cdot u)/(u \cdot u))u$, a multiple of u, not generally of y. For $y = (1, 1)$, $u = (1, 0)$ the projection $(1, 0)$ is not a multiple of y.

(j) **True.** A projection onto W belongs to W by its construction; equality with that projection places y in W.

(k) **True.** The set is the null space of the one-row matrix v^T for the fixed vector v, so is a subspace. This includes $v = 0$, when the set is all of $\mathbb{R}^n$.

(l) **False.** Their intersection is $\{0\}$, not empty. If x belongs to both, $x \cdot x = 0$, hence $x = 0$.

(m) **True.** For $i \neq j$, $(c_i v_i) \cdot (c_j v_j) = c_i c_j (v_i \cdot v_j) = 0$. Zero coefficients may introduce zero vectors, but pairwise orthogonality still holds.

(n) **False.** Orthonormal columns give $U^T U = I$. For $U = (1, 0)^T$, $UU^T = \text{diag}(1, 0) \neq I_2$. Squareness is needed to infer the reverse identity.

(o) **False.** Orthogonal columns need not have length one. The matrix $2I$ has orthogonal columns but $U^T U = 4I \neq I$.

(p) **True.** Square orthonormal-column U satisfies $U^T U = I$, hence $U^{-1} = U^T$ and $UU^T = I$. The latter identity says its rows are orthonormal.

(q) **True.** The projection and its residual are perpendicular, so the displayed identity is precisely Pythagoras.

(r) **False.** The minimizing input is $\widehat{x}$; $A\widehat{x}$ is the closest vector in $\operatorname{Col} A$. The inequality is valid, but it describes the output, not the least-squares solution itself.

(s) **False.** The normal equations are $A^TA\widehat{x} = A^Tb$. The inverse formula $\widehat{x} = (A^TA)^{-1}A^Tb$ requires independent columns. For $A = 0$, least-squares solutions exist but A^TA is not invertible.

Exercise 6.S.2: Inductively extend Pythagoras

For $p = 2$, orthonormality makes c_1v_1 perpendicular to c_2v_2; therefore $\|c_1v_1 + c_2v_2\|^2 = c_1^2\|v_1\|^2 + c_2^2\|v_2\|^2 = c_1^2 + c_2^2$. Assume the formula holds for $p-1$ terms. Let $w = \sum_{i=1}^{p-1} c_iv_i$. Since $w \cdot v_p = \sum_{i<p} c_i(v_i \cdot v_p) = 0$, Pythagoras gives

$$\|w + c_pv_p\|^2 = \|w\|^2 + c_p^2\|v_p\|^2 = \sum_{i=1}^{p-1} c_i^2 + c_p^2.$$

This completes induction. For real coefficients $c_i^2 = |c_i|^2$, so the result has exactly the form in the exercise.

Exercise 6.S.3: Bessel's inequality

Let $p = \sum_i (x \cdot v_i)v_i$, the projection of x onto the span of the orthonormal vectors. The residual $z = x - p$ is orthogonal to every v_i, because $z \cdot v_j = x \cdot v_j - (x \cdot v_j) = 0$. Thus $p \perp z$, and Exercise 2 yields

$$\|x\|^2 = \|p\|^2 + \|z\|^2 = \sum_i |x \cdot v_i|^2 + \|z\|^2 \geq \sum_i |x \cdot v_i|^2.$$

Nonnegativity of the residual's squared length is the entire source of the inequality. Equality holds precisely when x lies in the given span.

Exercise 6.S.4: An orthogonal map preserves an orthonormal basis

For each pair i, j,

$$(Uv_i) \cdot (Uv_j) = v_i^TU^TUv_j = v_i^Tv_j = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

Thus the n transformed vectors are orthonormal. They are independent: a zero combination, dotted with each transformed vector, forces each coefficient to vanish. An independent set of n vectors in $\mathbb{R}^n$ is a basis, proving the full claim.

Exercise 6.S.5: Preserving all inner products characterizes orthogonality

Apply the given identity to $x = e_i$, $y = e_j$. The vectors Ue_i, Ue_j are columns i, j of U, so their dot product is $e_i \cdot e_j$, equal to 1 for $i = j$ and zero otherwise. Hence the columns of U are orthonormal: $U^TU = I$. Since U is square, this is exactly the condition that U is orthogonal.

Exercise 6.S.6: Real eigenvalues of an orthogonal matrix

Suppose $Uv = \lambda v$ with $v \neq 0$ and real λ. Orthogonal matrices preserve length, giving $\|v\| = \|Uv\| = \|\lambda v\| = |\lambda|\|v\|$. Divide by $\|v\| > 0$ to obtain $|\lambda| = 1$. For a real number this says $\lambda = 1$ or -1. It does not claim every eigenvalue is real: planar rotations can have nonreal eigenvalues of modulus one.

Exercise 6.S.7: A Householder matrix is orthogonal

Put $P = uu^T$ with $u^T u = 1$. Then $P^T = P$ and $P^2 = u(u^T u)u^T = P$. For $Q = I - 2P$, symmetry gives $Q^T = Q$ and multiplication gives $Q^2 = I - 4P + 4P^2 = I$. Consequently $Q^T Q = Q^2 = I$, so Q is orthogonal. This proves length preservation as well as its being its own inverse.

Exercise 6.S.8: A length-preserving linear map is orthogonal

(a) Recover dot products from lengths.

By expansion, $2a \cdot b = \|a+b\|^2 - \|a\|^2 - \|b\|^2$. For $a = T(x)$, $b = T(y)$, linearity and length preservation give

$$2T(x) \cdot T(y) = \|T(x+y)\|^2 - \|T(x)\|^2 - \|T(y)\|^2 = \|x+y\|^2 - \|x\|^2 - \|y\|^2 = 2x \cdot y.$$

In particular perpendicular inputs have perpendicular outputs.

(b) Apply this to the standard basis.

Each $T(e_i)$ has length one, and distinct $T(e_i), T(e_j)$ are perpendicular by part (a). These vectors are the columns of the square standard matrix, so that matrix has orthonormal columns and is orthogonal.

Exercise 6.S.9: Best approximation without orthogonalizing the generators

Set $A = [u\ v]$. The problem is to minimize $\|z - Ac\|$ for $c = (x_1, x_2)^T$. Solve the two normal equations

$$\begin{bmatrix} u \cdot u & u \cdot v \\ u \cdot v & v \cdot v \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} u \cdot z \\ v \cdot z \end{bmatrix}.$$

Their coefficient matrix is invertible because u, v are independent: for nonzero c, $c^T A^T A c = \|Ac\|^2 > 0$. The desired closest vector is $x_1 u + x_2 v$. These equations explicitly account for the nonzero cross term $u \cdot v$; using the separate orthogonal-projection coefficients would be wrong when the generators are not perpendicular.

Exercise 6.S.10: Scale the data, scale the least-squares solution

Independent columns imply $A^T A$ is invertible and the solution is unique. Thus replacing b by cb changes the solution to

$$(A^T A)^{-1} A^T (cb) = c(A^T A)^{-1} A^T b = c\widehat{x}.$$

One can also multiply the original normal equations by c and use uniqueness. The assertion even holds for $c = 0$, though the question only requests nonzero scalars.

Exercise 6.S.11: Fit three parallel inconsistent planes

Write $t = x - 2y + 5z$ and $\mu = (a + b + c)/3$. The squared residual is

$$(t-a)^2 + (t-b)^2 + (t-c)^2 = 3(t-\mu)^2 + (a-\mu)^2 + (b-\mu)^2 + (c-\mu)^2.$$

The equality follows by expanding; the mixed terms cancel because $(a-\mu) + (b-\mu) + (c-\mu) = 0$. The last three terms are constant, so the minimum occurs exactly at $t = \mu$. Every such t is attainable, for instance by taking $x = t, y = z = 0$. Therefore all least-squares solutions, and only those, form the plane

$$\boxed{x - 2y + 5z = (a + b + c)/3}.$$

The least-squares input is nonunique because this equation leaves two free variables, even though the best fitted output is unique.

Exercise 6.S.12: A Rayleigh quotient as a one-variable least-squares fit

Assume the proposed approximate eigenvector v is nonzero. Treat $v\lambda = Av$ as a least-squares system with one unknown. Its normal equation is $(v^T v)\lambda = v^T A v$. Since $v^T v = \|v\|^2 > 0$, its unique solution is

$$\boxed{\widehat{\lambda} = \frac{v^T A v}{v^T v}}.$$

This is the Rayleigh quotient. If $Av = \lambda_0 v$ exactly, the quotient returns λ_0. For $v = 0$ every scalar gives the same zero residual, so there would be no meaningful eigenvalue estimate; nonzeroness is essential.

Exercise 6.S.13: Identify the four fundamental subspaces

(a) If $u \in \operatorname{Nul} A$, each row r_i^T of A satisfies $r_i^T u = 0$. A linear combination of those rows is still perpendicular to u. Thus every member of $\operatorname{Row} A$ is perpendicular to every member of $\operatorname{Nul} A$, proving the stated containment.

(b) If $\operatorname{rank} A = r$, rank–nullity gives $\dim \operatorname{Nul} A = n - r$. A subspace of $\mathbb{R}^n$ and its orthogonal complement have dimensions adding to n, so $\dim(\operatorname{Nul} A)^\perp = r$. The row space also has dimension r. A contained subspace with the same finite dimension must be the whole space, so $\operatorname{Row} A = (\operatorname{Nul} A)^\perp$.

(c) Apply this result to A^T. Since $\operatorname{Row}(A^T) = \operatorname{Col} A$, it gives $\operatorname{Col} A = (\operatorname{Nul} A^T)^\perp$, now inside the ambient space $\mathbb{R}^m$.

Exercise 6.S.14: The consistency alternative in orthogonal-complement language

The system $Ax = b$ is consistent if and only if $b \in \operatorname{Col} A$. By Exercise 13(c), this is equivalent to $b \in (\operatorname{Nul} A^T)^\perp$, which says b is orthogonal to every solution of $A^T y = 0$. This proves both directions of the equivalence. Orthogonality to just one selected null vector would not in general be sufficient.

Exercise 6.S.15: A real triangular Schur factorization gives real eigenvalues

The factorization $A = URU^T$ with orthogonal U makes A similar to R. Similar matrices have the same characteristic polynomial. Since R is upper triangular, $\det(R - tI) = \prod_{i=1}^{n}(r_{ii} - t)$. Its n roots, counted with multiplicity, are the real diagonal entries r_{ii}. Hence A has n real eigenvalues counting multiplicities. The restriction to a real, genuinely triangular R matters; real Schur forms for matrices with nonreal eigenvalues use 2×2 blocks instead.

Exercise 6.S.16: The first step in constructing a real Schur factorization

(a) Since $Ue_1 = u_1$ and $Au_1 = \lambda_1 u_1$,

$$(U^T AU)e_1 = U^T Au_1 = \lambda_1 U^T u_1 = \lambda_1 e_1.$$

This is precisely the assertion about the first column. Here $U^T u_1 = e_1$ follows from the orthonormality of the columns.

(b) The block upper triangular form gives

$$\det(U^T AU - tI) = (\lambda_1 - t)\det(A_1 - tI).$$

Similarity makes this equal to $\det(A - tI)$, whose roots, with multiplicity, are the listed $\lambda_1, \ldots, \lambda_n$. Cancelling the polynomial factor $\lambda_1 - t$ leaves the roots $\lambda_2, \ldots, \lambda_n$ for A_1. This argument also handles repeated values of λ_1: only one copy is removed from the multiplicity list.

Exercise 6.S.17: Check a condition-number bound for perturbed data

Use the Euclidean norm throughout and $\kappa_2(A)$ as the condition number. Exact rational solution of the two systems gives

$$x = (3.94, .49)^T, \qquad \Delta x = (-1.04, 1.51)^T.$$

For example $A^{-1} = \begin{bmatrix} -110 & 310 \\ 160 & -450 \end{bmatrix}$, so multiplying by b and Δb verifies both vectors. The measured ratios and bound are

$$\frac{\|\Delta x\|_2}{\|x\|_2} \approx .461796, \quad \frac{\|\Delta b\|_2}{\|b\|_2} \approx .000154792, \quad \kappa_2(A) \approx 3362.9997.$$

Their product on the right is about .520567, which exceeds .461796. Thus the asserted inequality holds for this example, despite the large relative solution change.

Exercise 6.S.18: The same perturbation with a different original solution

The same inverse and the new right side give

$$x = (-491.17, 713.15)^T, \qquad \Delta x = (-1.04, 1.51)^T.$$

The perturbation is unchanged because A and Δb are unchanged. However the relative ratios use the new original vector:

$$\frac{\|\Delta x\|_2}{\|x\|_2} \approx .00211738, \qquad \frac{\|\Delta b\|_2}{\|b\|_2} \approx .00211778.$$

Multiplying the latter by 3362.9997 gives about 7.12211, much larger than .00211738. The bound is valid but far from attained. A condition number describes worst-case amplification, not the amplification of every particular perturbation.

Exercise 6.S.19: A small change along a relatively harmless direction

Exact solution of the two printed systems gives

$$x = (-8.642, -1730.054, 2368.716, -844.866)^T,$$

$$\Delta x = (.000030, .000140, -.000155, .000059)^T.$$

These terminating decimals are exact for the printed decimal data. Direct multiplication by the given integer A yields the stated b and Δb, respectively, providing an independent check. Numerically,

$$\kappa_2(A) \approx 23682.9279, \quad \frac{\|\Delta x\|_2}{\|x\|_2} \approx 7.17780 \cdot 10^{-8}, \quad \frac{\|\Delta b\|_2}{\|b\|_2} \approx .000283204.$$

The right-hand bound is approximately 6.70710, far larger than the observed $7.18 \cdot 10^{-8}$. Therefore the inequality holds. The original solution is large, while the particular change in it is tiny.

Exercise 6.S.20: A perturbation close to worst-case amplification

With the new right side and perturbation, exact rational arithmetic gives

$$x = (2.541, 1.460, 1.174, -.101)^T, \qquad \Delta x = (-.002322, -.464848, .636450, -.227007)^T.$$

Multiplication checks $Ax = b$ and $A\Delta x = \Delta b$. The condition number is the same as in Exercise 19, since A is unchanged. The ratios are

$$\frac{\|\Delta x\|_2}{\|x\|_2} \approx .259665, \qquad \frac{\|\Delta b\|_2}{\|b\|_2} \approx 1.09702 \cdot 10^{-5}.$$

The right-hand bound is .259806, slightly larger than .259665. Thus the inequality holds and is nearly sharp for this choice. All four experiments can be reproduced using exact SymPy solves for x and Δx, then NumPy's `linalg.cond(A)` and Euclidean vector norms. Matching the norm used for the condition number and the relative changes is essential.